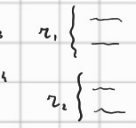
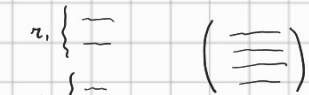
due rette che non si intersecano sono:

- PARALLELE se esiste un piano che le contiene $rk A = 2$ $rk A|B = 3$
- SGHEMBE se NON " " " " " $rk A = 3$ $rk A|B = 4$

r_1, r_2 



Se r_1, r_2 sono sghembe, esistono due piani π_1, π_2 paralleli t.c. $r_1 \subseteq \pi_1$ e $r_2 \subseteq \pi_2$

esempio:

$$r_1: \begin{cases} x - 2y + 3z = 0 \\ 2x - z + 1 = 0 \end{cases} \qquad r_2: \begin{cases} x - y = 0 \\ x + y + z = 0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 2 & 0 & -1 & -1 \\ 1 & -2 & 3 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & -1 & -1 \\ 1 & -2 & 3 & 0 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 \\ 1 & -2 & 3 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & -2 & -1 \\ 0 & -3 & 2 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & -2 & -1 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -6 & -3 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & -2 & -1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & -3 & 0 & -1 \\ 0 & 0 & -2 & -1 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -3 & 0 & -1 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 0 & 2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1/3 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1/3 - 1/2 \\ 0 & 1 & 0 & 1/3 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{5}{6} \\ 0 & 1 & 0 & 1/3 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

→ le due rette sono sghembe

$R_3 \to R_3 - \frac{1}{2} R_2$

$R_4 \to R_4 + \frac{1}{2} R_2$

$R_4 \to R_4 + \frac{13}{5} R_3$

Come troviamo ora il vettore direzionale di r_1 e dei piani che stiamo cercando.

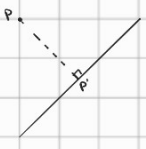
Il PRODOTTO VETTORIALE non ha la proprietà associativa

$$((2,-4,1) \wedge (1,-2,3)) \wedge ((1,-1,0) \wedge (1,1,1)) =$$

$$= (-2, -(7), -1) \wedge (-1, -(1), 2)$$

$$= (-18, -(-8), -5) = -18x + 8y - 5z = d \quad \Rightarrow \quad \pi_1: -18x + 8y - 5z = 7, \quad \pi_2: -18x + 8y - 5z = 0$$

fascio di piani



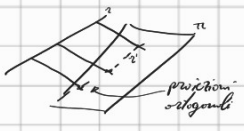


$P(1,2,0)$

$$r: \begin{cases} x + 2y - z = 0 \\ x + y = 1 \end{cases}$$

$$r: \begin{cases} x = t + 1 \\ y = -2t + 2 \\ z = -t \end{cases} \quad t \in \mathbb{R}$$

Proiezione di una retta su un piano



$$r \begin{cases} x + y + z = 3 \\ x - y - z = 2 \end{cases}$$

$\pi: x + 2y + 3z = 4$

Fascio di piani per r

$$\lambda(x + y + z - 3) + \mu(x - y - z - 2) = 0$$

$$(\lambda - \mu, \lambda - \mu, \lambda - \mu)$$

$$\lambda - \mu + 2(\lambda - \mu) + 3(\lambda - \mu) = 0$$

$$6\lambda - 4\mu = 0$$

posso prendere qualunque valore come $\lambda = 2$ $\mu = 3$

$5x - y - z = 12$

$$r': \begin{cases} 5x - y - z = 12 \\ x + 2y + 3z = 4 \end{cases}$$